

PPGEE2249 Aprendizado de Máquina
Assignment 3
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► Question 1

Train and evaluate a Feedforward Neural Network for a multiclass classification task. Split your data into a training set, a validation set (used to define the stopping criterion - e.g., number of epochs), and a test set for the final evaluation of the model. The output layer should use one Softmax neuron per class, and the training objective should be the minimization of cross-entropy loss. Describe your dataset, the network architecture (hidden units, layer dimensions, and number of layers), and discuss the overall performance (data splitting strategy, classification accuracy, and confusion matrix).

- $0 \leq l \leq L$ (layer index)
- $1 \leq t \leq T$ (input index)
- $\mathbf{x}_t \in \mathbb{R}^{d \times 1}$ (input)
- $\mathbf{x}_t^* = [1 \ \mathbf{x}_t^\top]^\top \in \mathbb{R}^{(d+1) \times 1}$ (augmented input)
- $\phi : \mathbb{R}^{(d+1) \times 1} \rightarrow \mathbb{R}^{(d+1) \times 1}$ (activation function)
- $\mathbf{W}_t^l \in \mathbb{R}^{(d+1) \times (d+1)}$ (layer weights)
- $\mathbf{h}_t^l \in \mathbb{R}^{(d+1) \times 1}$ (layer output)
- $\text{softmax} : \mathbb{R}^{c \times 1} \rightarrow \mathbb{R}^{c \times 1}$ s.t.

$$\text{softmax}([u_1 \ u_2 \ \cdots \ u_c]^\top) = [\exp(u_1) \ \exp(u_2) \ \cdots \ \exp(u_c)]^\top / \sum_k \exp(u_k)$$

- $\mathbf{V}_t \in \mathbb{R}^{c \times (d+1)}$ (output weights)
- $\mathbf{y}_t \in \mathbb{R}^{c \times 1}$ (output)
- $\mathbf{r}_t \in \mathbb{R}^{c \times 1}$ (label)
- L (loss/error function)

$$\mathbf{h}_t^l = \begin{cases} \mathbf{x}_t^*, & l = 0 \\ \phi(\mathbf{W}_t^l \mathbf{h}_t^{l-1}), & l > 0 \end{cases} \quad (1)$$

$$\mathbf{y}_t = \text{softmax}(\mathbf{V}_t \mathbf{h}_t^L) \quad (2)$$

$$L_t(\mathbf{W}, \mathbf{V}) = - \sum_{\tau \leq t} \sum_i r_{\tau i} \log y_{\tau i} \quad (3)$$

Problem: change $\mathbf{W}_t^l, \mathbf{V}_t$ to minimize L_t .

Solution: apply the chain rule of the derivative.

$$L_t(\mathbf{W}, \mathbf{V}) = - \sum_{\tau \leq t} \sum_i r_{\tau i} \log y_{\tau i} \quad (4)$$

$$= - \sum_{\tau \leq t} \sum_i r_{\tau i} \log (\text{softmax}(\mathbf{v}_t^i \mathbf{h}_t^L)) \quad (5)$$

$$= - \sum_{\tau \leq t} \sum_i r_{\tau i} \log \left(\text{softmax} \left(\sum_{j=0}^d v_{tij} h_{tLj} \right) \right) \quad (6)$$

$$= - \sum_{\tau \leq t} \sum_i r_{\tau i} \log \left(\text{softmax} \left(\sum_{j=0}^d v_{tij} \phi^{(L)}(h_{tLj}) \right) \right) \quad (7)$$

► Question 2

Train and evaluate a Decision Tree on the same classification problem chosen in Question 1. Present the resulting tree (depth, rule-based description) and compare its performance with the results obtained in Question 1.

► Question 3

Perform an experiment testing different kernel functions and hyperparameter configurations to obtain the best SVM classifier for a binary classification task, using a dataset of your choice. Don't forget to discuss the results.
